

Authored by : Arpit Raj , Lnmiit jaipur

Chapter 7 — STDDEV()

7.1 Introduction

- Standard deviation.

7.2 Why Standard Deviation Matters

7.3 Population vs Sample

Two versions exist.

- STDDEV_POP
- STDDEV_SAMP

Population

STDDEV_POP()

Population standard deviation.

Use when the table contains every data point.

Example: You have salaries of all employees.

Difference: divide by N

Sample

STDDEV_SAMP()

Sample standard deviation.

Use when you only have a sample.

Example: 100 employees chosen out of 20,000 employees.

Difference: divide by N-1

Sample standard deviation is usually slightly larger.

7.4 Syntax

```
SELECT STDDEV(salary) FROM Employee;
```

or

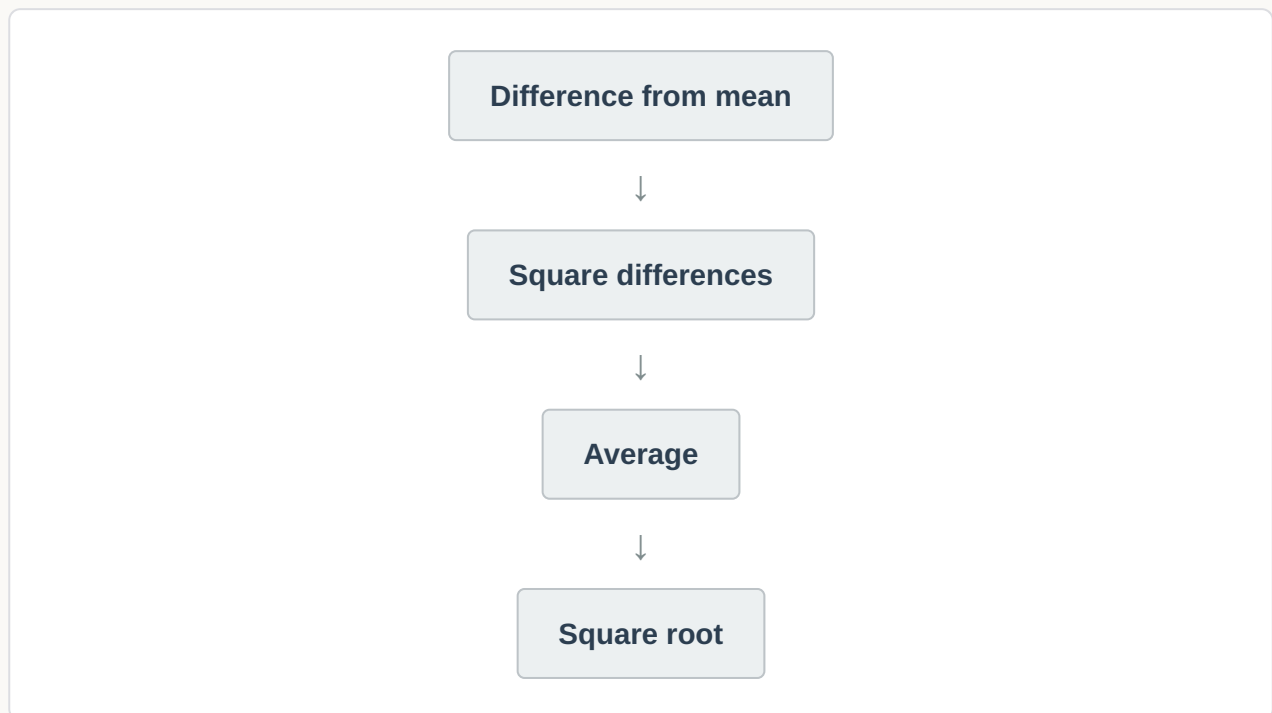
```
SELECT STDDEV_POP(salary) FROM Employee;
```

or

```
SELECT STDDEV_SAMP(salary) FROM Employee;
```

7.5 Mathematical Interpretation

Formula (conceptually)



7.6 Business Examples

- Salary variation
- Sales consistency
- Exam scores

7.7 Practice Problems

Chapter 8 — VARIANCE()

8.1 Introduction

- Variance.

Variance measures spread just like standard deviation.

Difference



Formula: $\text{Variance} = \text{STDDEV}^2$

8.2 Population vs Sample

Population

- VAR_POP
- VAR_POP()

Sample

- VAR_SAMP
- VAR_SAMP()

8.3 Syntax

```
SELECT VAR_POP(salary) FROM Employee;
```

or

```
SELECT VAR_SAMP(salary) FROM Employee;
```

8.4 Relationship with STDDEV

8.5 Practical Examples

8.6 Practice Problems

Chapter 9 — MEDIAN()

9.1 Introduction

- Middle value.

Median is the middle value after sorting.

Unlike AVG,
Median is not affected much by outliers.

9.2 Mathematical Definition

9.3 Odd Number of Rows

9.4 Even Number of Rows

9.5 Vendor Support

MySQL

- No built-in MEDIAN.
- Need window functions.

9.6 Manual Median Calculation

- Using window functions

Manual Median

Later with window functions.

Concept

- ROW_NUMBER()
- COUNT()
- Middle row

9.7 Business Applications

9.8 Practice Problems

Chapter 10 — MODE()

10.1 Introduction

- Most frequent value.

Mode is the most frequently occurring value.

Example

10, 20, 20, 30, 20, 40

Mode 20

10.2 Statistical Meaning

Highest frequency.

10.3 Vendor Support

10.4 Manual Mode Calculation

- Using GROUP BY

10.5 Multiple Modes

10.6 Practical Examples

- Most common city
- Most purchased product

ProductABACAB

```
SELECT product,  
       COUNT(*) AS frequency  
FROM Products  
GROUP BY product  
ORDER BY frequency DESC  
LIMIT 1;
```

Output: A

- Most common grade

```
SELECT grade,  
       COUNT(*)  
FROM Student  
GROUP BY grade  
ORDER BY COUNT(*) DESC  
LIMIT 1;
```